INCH-POUND

MIL-PRF-3098/108E
w/AMENDMENT 1
7 October 2016

SUPERSEDING
MIL-PRF-3098/108E
6 March 2013

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR128/U

INACTIVE FOR NEW DESIGN
AFTER 22 JUNE 2016

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.

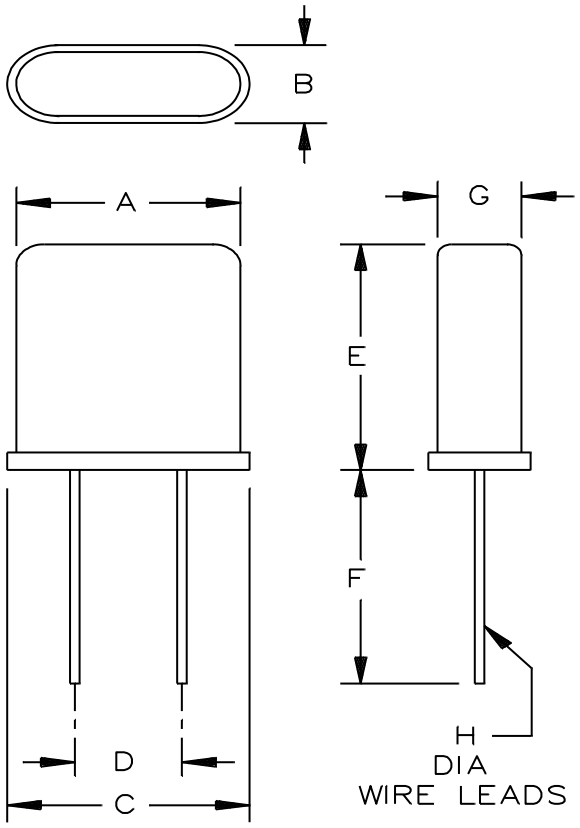


FIGURE 1. Crystal unit - CR128/U.

AMSC N/A

FSC 5955

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR128/U, crystal unit - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 MHz to 61 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Mode of oscillation: Third mechanical overtone.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Operable temperature range: +105°C to +125°C, inclusive.

Frequency tolerance (operating temperature range): ±50 ppm.

Equivalent resistance: 40 ohms, maximum.

Qualification: Not required. First article testing shall be performed according to the qualification table of MIL-PRF-3098.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Vibration: MIL-STD-202-204, test condition A; 0.015 inch (0.03 inch total excursion) 10 Hz to 55 Hz; or an amplitude to produce 10 gravity units, 55 Hz to 500 Hz, whichever is the limiting value.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Thermal shock: ±50 ppm.

Aging:

Frequency change permitted: ±10 ppm.

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Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204

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Custodians:
Army - CR
Navy - EC
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2016-039)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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